LAB PROGRAMS (6-10)

COMPUTER VISION-ITA-0505

SLOT-D

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QUESTION-6

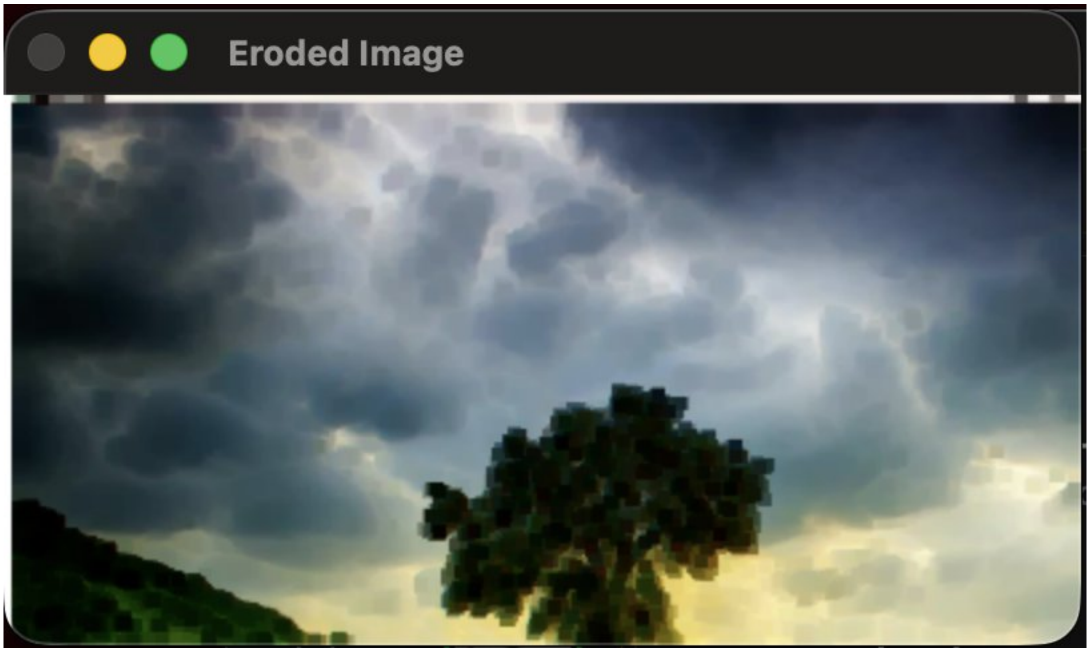
Perform basic Image Handling and processing operations on the image is to read an image in python and Convert an Image to erode using Erode function.

PYTHON CODE:

```
import cv2
# Read the image
image = cv2.imread("sample.jpg")
# Check if the image is loaded
if image is None:
    print("Error: Image not found!")
    exit()
# Convert image to grayscale
gray_image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
# Display original image
cv2.imshow("Original Image", image)
# Display grayscale image
cv2.imshow("Gray-scale Image", gray_image)

# Wait for a key press
```

```
cv2.waitKey(0)
# Close all windows
cv2.destroyAllWindows()
```

OUTPUT:



QUESTION-7:

Perform basic video processing operations on the captured video. Read captured video in python and display the video, in slow motion and in fast motion.

PYTHON CODE:

```
import cv2
# Read the image
image = cv2.imread("sample.jpg")
 # Check if the image is loaded
if image is None:
```

```
    print("Error: Image not found!")
    exit()
# Apply Gaussian Blur
blur_image = cv2.GaussianBlur(image, (15, 15), 0)
# Display original image
cv2.imshow("Original Image", image)
# Display blurred image
cv2.imshow("Gaussian Blur Image", blur_image)
# Wait until a key is pressed
cv2.waitKey(0)
# Close all windows
cv2.destroyAllWindows()
```

OUTPUT:


Fast Motion Video
Type to Siri
TATA IPL
FINAL


Normal Video
Type to Siri
TATA IPL
FINAL

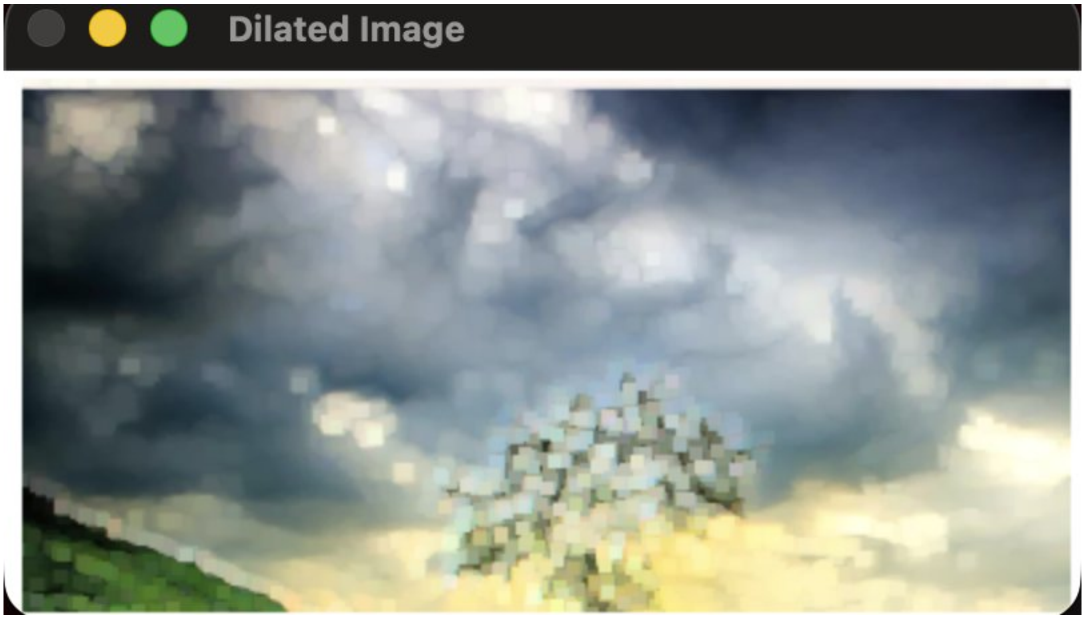
QUESTION-8:

Perform basic Image Handling and processing operations on the image is to read an image in python and Dilate an Image using Dilate function

PYTHON CODE:

```
import cv2
# Read the image
image = cv2.imread("sample.jpg")
# Check if image is loaded
if image is None:
    print("Error: Image not found!")
    exit()
# Convert to grayscale
gray_image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
# Apply Canny edge detection
edges = cv2.Canny(gray_image, 100, 200)
# Display images
cv2.imshow("Original Image", image)
cv2.imshow("Edge Detected Image", edges)
# Wait for key press
cv2.waitKey(0)
# Close all windows
cv2.destroyAllWindows()
```

OUTPUT:



QUESTION-9:

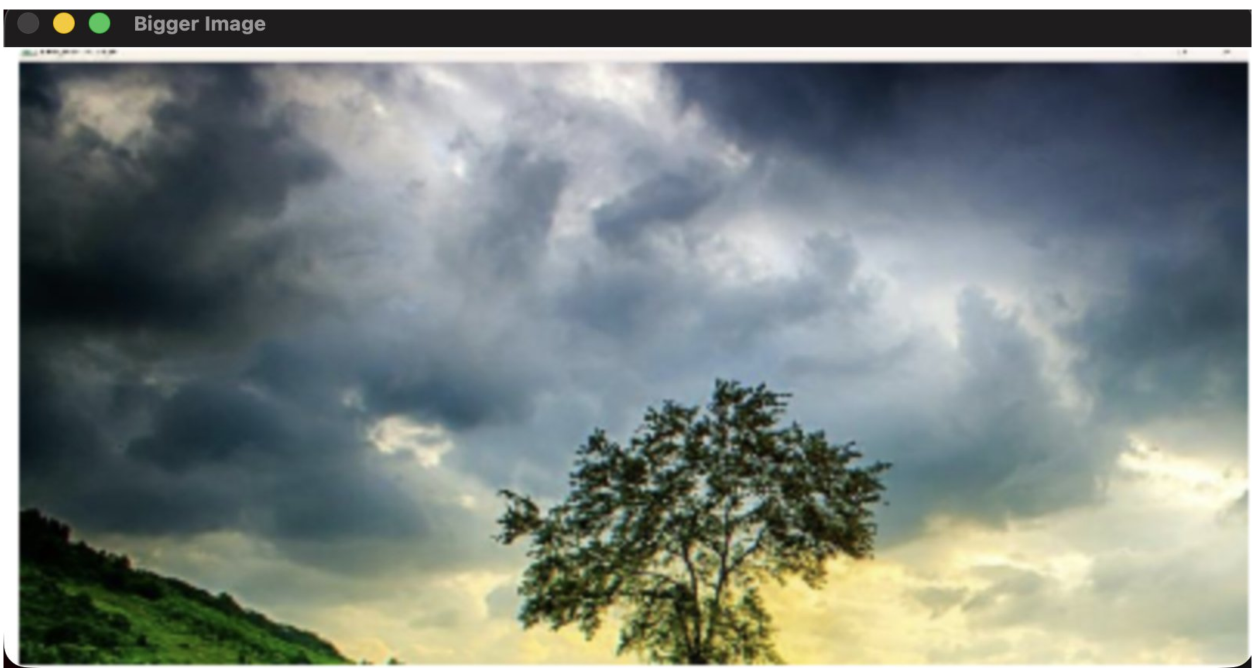
Implement the image scaling techniques to resize the images to both bigger and small sizes

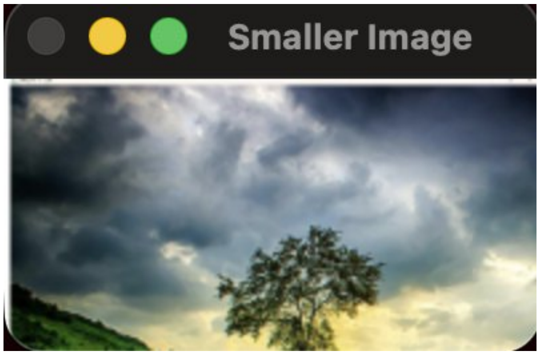
PYTHON CODE:

```
import cv2
# Read the image
image = cv2.imread("sample.jpg")
# Check if image is loaded
if image is None:
    print ("Error: Image not found!")
    exit ()
# Convert to grayscale
gray_image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
# Apply Histogram Equalization
```

```
equalized_image = cv2.equalizeHist(gray_image)
# Display images
cv2.imshow("Original Grayscale Image", gray_image)
cv2.imshow("Histogram Equalized Image", equalized_image)
# Wait until a key is pressed
cv2.waitKey(0)
# Close all windows
cv2.destroyAllWindows()
```

OUTPUT:





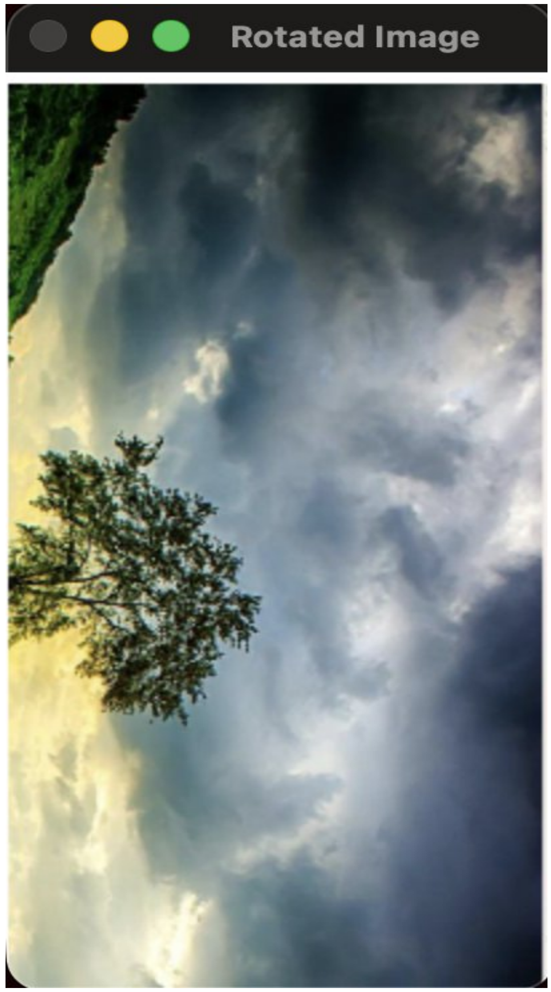
Q U E S T I O N - 1 0 :
Perform a 90-degree rotation clockwise along the y-axis for the given image.

PYTHON CODE:

```
import cv2

import matplotlib.pyplot as plt

def analyze_histogram(image_path):

    # Read the image

    image = cv2.imread(image_path)

    # Check if image is loaded

    if image is None:

        print("Error: Image not found!")

        return

    # Display the original image

    cv2.imshow("Original Image", image)

    # BGR color channels

    colors = ('b', 'g', 'r')

    # Create histogram plot
```

```
    plt.figure(figsize=(8, 5))
    for i, color in enumerate(colors):
        histogram = cv2.calcHist([image], [i], None, [256], [0, 256])
        plt.plot(histogram, color=color, label=f"{color.upper()} Channel")
    plt.title("Color Histogram Analysis")
    plt.xlabel("Pixel Intensity")
    plt.ylabel("Number of Pixels")
    plt.xlim([0, 256])
    plt.legend()
    # Show histogram
    plt.show()
    # Wait for key press
    cv2.waitKey(0)
    cv2.destroyAllWindows()
# Call the function
analyze_histogram("sample.jpg")
```

OUTPUT:


Rotated Image